

Copper in the USA: A Production, Price, and Renewable Energy Lens

A Time Series and Forecasting Analysis

Prepared for Environmental Data Analysis and Visualization

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Project Focus

This report independently studies US copper production, global prices, and US-native installed renewable energy capacity, and also looks at the relationships between them. Forecasts have been made for all the variables.

Introduction

Copper is a crucial component of the clean energy transition due to its high durability, conductivity, and recyclability (S&P Global, n.d.). Copper parts and wires are responsible for transmitting the energy required for clean power, EVs, AI data centres and the utilities (electricity) grid. The purpose of our project is to analyse how the United States, a global economic and technological leader who is heavily relying on copper imports among other critical minerals. How has copper production in the US changed over time, and how much of it has been driven by the influence of installed capacity of renewable energy infrastructure and fluctuations in global copper price (traded internationally in USD)?

Using verifiable official data 1990–2024 data from United States Geological Survey (USGS), the International Energy Agency (IEA) and the World Bank, we examine copper across three angles: supply, demand and price. The core of our work is time-series analysis. We have tested each series for the presence of a trend using Mann-Kendall and Theil-Sen's slope, checked stationarity using ADF and KPSS, and performed forecasts till 2030 using ARIMA (since our data is annual, we have avoided SARIMA). Our results have been benchmarked against naive models with validation using back-testing and reasonably wide uncertainty bands given the unpredictability of commercial and economic variables such as the ones we have chosen.

US production shows a slow, price-driven decline (the increase in global copper prices explains more of the decline in production). The decline in production is actually counter-intuitive given the exponential increase